

Digital serial port PIR Sensor YS312

General Description

YS312 is a type of PIR sensor which integrates AD chip and sensitive element in a hermetically sealed package. The sensitive element transmits the sensed human body movement signal to the AD chip, which outputs 16-bit digital signals through collection, filtering, etc., and realizes related functions through the peripheral circuit MCU.

Features

- 16-bit digital signal output
- Single-wire serial data
- Low power consumption

Applications

Consumer electronics applications:

- Toy
- Digital photo frame, Doorbell
- TV, refrigerator, Air conditioner

Smart home, security applications:

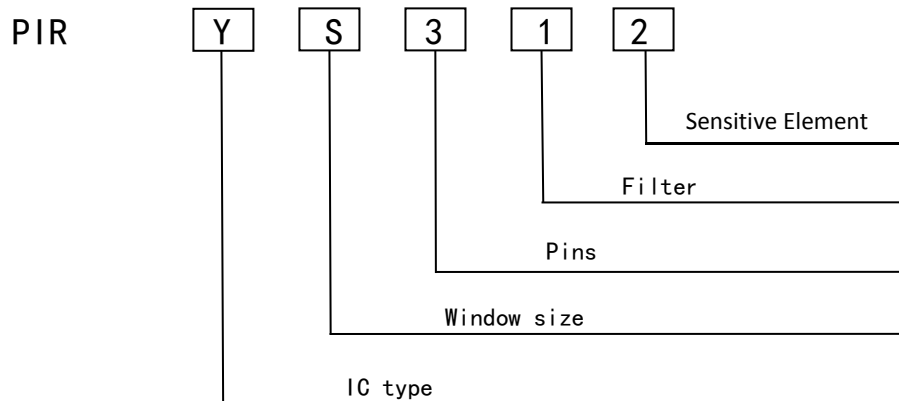
- USB alarm
- Intrusion detection
- Network camera
- LAN monitor
- Private alarm
- Car alarm system

Lighting application:

- Automatic lighting of indoors, courtyards, corridors, staircase lights, etc.



Product Type Series



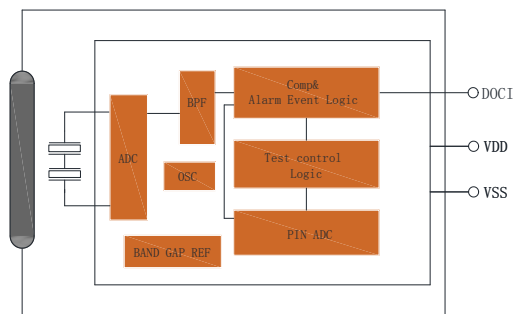
Technical data

1. Absolute Maximum Ratings

Exceeding the values in the table below will cause permanent damage to the device.

Parameter	Symbol	Min	Max	Unit	Remarks
Operating Temp	T _{ST}	-30	70	°C	
Current into any pin	I _{nto}	-100	100	mA	
Storage Temp	T _{ST}	-40	80	°C	

1.1 Internal schematic diagram



2. Operating Conditions (T=25°C, V_{dd}=3V, unless stated otherwise)

Parameter	Symbol	Min	Type	Max	Unit	Remarks
Regular voltage	V _{DD}	2.2	3.3	5.5	V	
Operation current	I _{DD}	5	10	20	μA	
ADC Resolution			16		Bits	
Detective Wavelength	λ	5		14	um	
LPF cutoff frequency				7	Hz	
HPF cutoff frequency				0.44	Hz	

Data format

1. Bandpass filter

The second-order low-pass filter and the third-order high-pass filter are cascaded to form a band-pass filter, and the corresponding band-pass frequency range is 0.44Hz to 7Hz respectively.

2. Serial data read timing

The DOCI timing diagram is shown in Figure 1. The dashed line represents the PIR drive, while the solid line represents the microcontroller drive. The timing time and data bits are shown in the table below:

tS	Starting signal time	100us~5ms	During continuous reading, tS should remain constant; otherwise, the data is prone to anomalies.
tL	Pull down time	4us~8us	
tH	Pull up time	4us~8us	
tB	Waiting for data stabilization time	4us~8us	
tR	PIR data update time	≈16ms	
1	High level, data 1		
0	Low level, data 0		

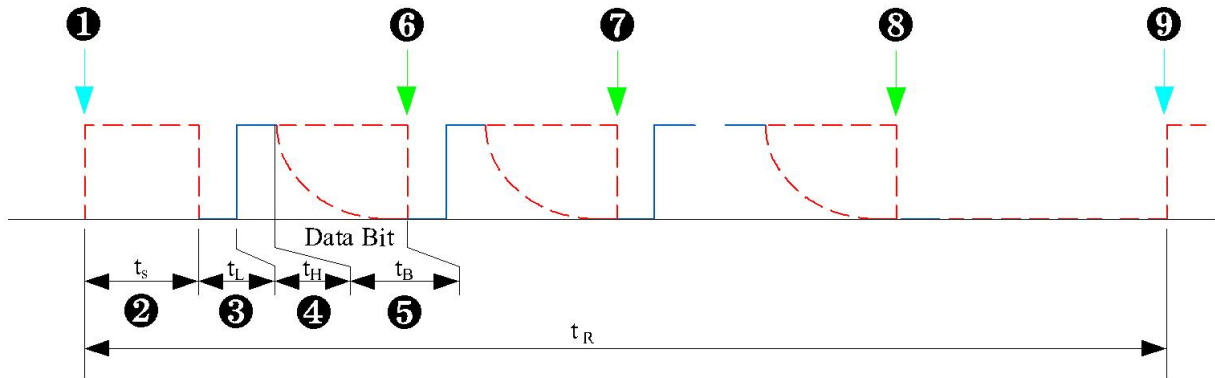


Figure 1

- ① When the DOCI line is idle, PIR completes data updates within 16ms, and PIR will pull up the DOCI line. If the DOCI line is idle for more than 16ms, the microcontroller needs to raise the DOCI line and release it.
- ② After PIR raises the DOCI line, maintain at least 100us~5ms.
- ③ The microcontroller pulls down the DOCI line for 4us to 8us.
- ④ After the microcontroller raises the DOCI line by 4us to 8us, release the DOCI line.

- ⑤ Pull down/up the DOCI line of the PIR, output the level state of the high bit data, and wait for the level state to stabilize with a delay of 4us to 8us.
- ⑥ The microcontroller samples the level state. A high level represents data '1', and a low level represents data '0'. The high - order data is read first.
- ⑦ Repeat③④⑤⑥ until all 19 bits of data are read out.
- ⑧ After reading 19 bits of data, the microcontroller lowers the DOCI line and releases it (refer to the note).
- ⑨ Repeat ① and start a new read cycle.

Note:

Release: The microcontroller switches to input mode without opening the up/down resistor.

Idle: The microcontroller releases the DOCI line and the DOCI line is at a low level.

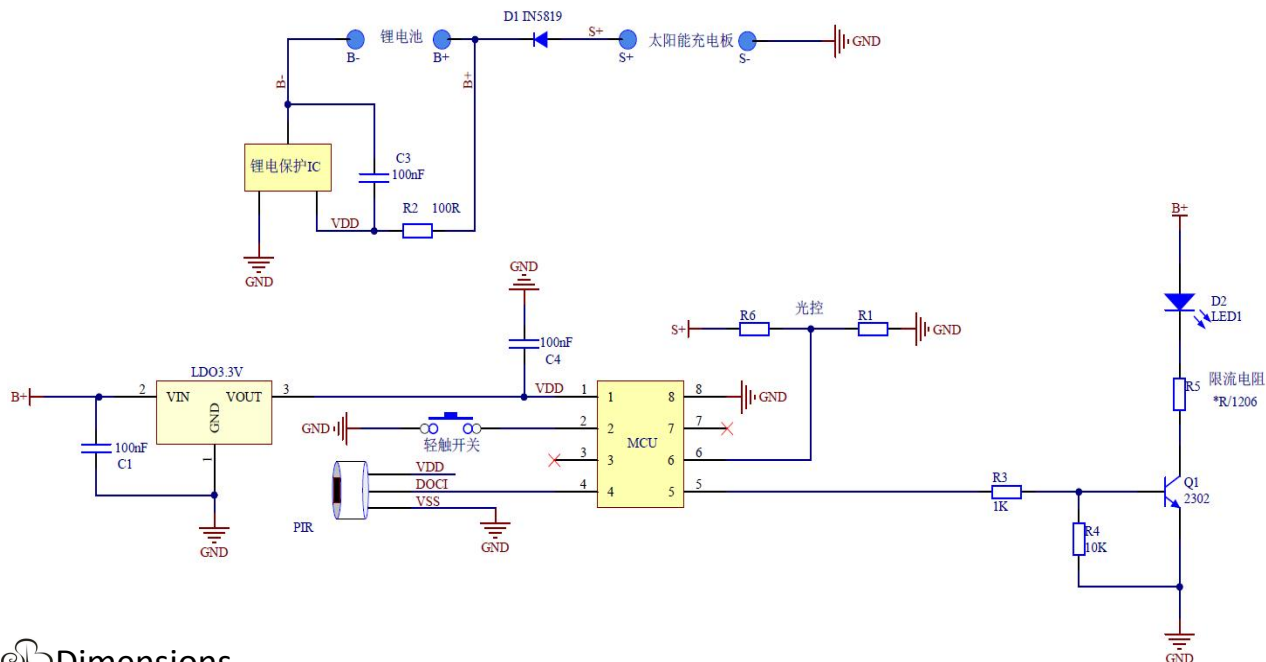
Microcontroller pull up/down: The microcontroller first sets the output high/low level, and then switches to output mode to prevent level changes caused by setting the output high/low level after switching to output mode.

3.Data Format

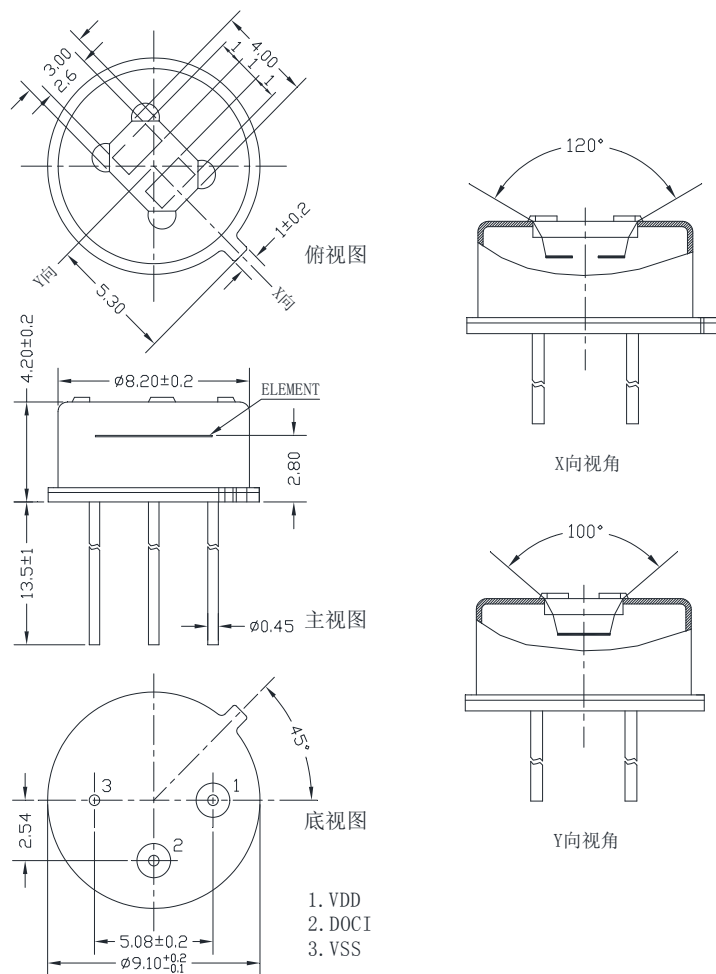
The DOCI interface outputs the value of the high pass filter. The DOCI output data is shown in the table below, with a total of 19 bits. The header code B18, B17 are fixed values of '1' and '0', and the tail code B0 is fixed value of '0'.

数值	头码		16位数据																尾码
	B18	B17	B16	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
32767	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
32766			0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	
.....																			
4			0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	
3			0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	
2			0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	
1			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
0			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
-1			1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
-2			1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	
-3			1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	
-4			1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	
.....																			
-32767			1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
-32768			1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Application Circuit



Dimensions



Sensor External View (UNIT: mm)

Reliable Test

No.	Test item	Test condition	Test equipment	Remarks
1	Moisture resistance	Temperature 35 °C, relative humidity 95% RH, for 500H	Temp&Humi Chamber	After testing, the sensor placed 3H in the natural environment. then test the electrical performance. Standard: 1.No obvious change in appearance. 2.The electrical parameter changes within 15%.
2	Low temperature storage	Temperature -40 °C, for 500H	Temp&Humi Chamber	
3	High temperature storage	Temperature 80 °C, time 500H	Temp&Humi Chamber	
4	High and low temperature impact	-40 °C, 1H / room temperature, 1H / 40 °C, 1H for 10 cycles	Thermal shock test chamber	
5	Solder resistance	Temperature 260±5 °C, 10S immersion depth 3MM	Tin stove	
6	vibration	Frequency 10-55Hz, vibration time 2H at 3 axis direction	Vibration test bench	No bubble
7	seal	Pressure 0.1MPa for 1Min	Vacuum tester	

Caution

Circuit design

1. The connection between PIR Sensor and other devices should be as short as possible. On the multi-panel or multi-layer board, try not to trace the line below the connection, especially no large current.

2. The circuit of the PIR human sensing part should be made in one single PCB board to avoid interference. If done on the same board, the PIR body sensing part of the circuit should be isolated and have a separate ground; only connect the other circuits through the positive, negative and output three lines.

3. VDD of the PIR Sensor should connect to ground with 100NF capacitor as close as possible.

Soldering Requirment

Soldering requirements

Manual soldering requirements

1. For manual soldering, soldering must be completed within 3 seconds and the soldering iron temperature is below 350°C。

Reflow soldering:

1. The wave soldering temperature is set at $260 \pm 5^{\circ}\text{C}$ and the speed $1500 \pm 300\text{mm/min}$.

2. The furnace temperature should not exceed 260°C , and the time should not exceed 3S.
(Note: The induction performance will be degraded or the function will be invalid by high temperature.)

3. During wave soldering, the PIR should not be tightly close to the PCB. It is recommended that 1mm or more distance, to prevent short-circuited.

4. Wipe the widow with dry cloth or absolute ethanol if there is stain.

Application

1. The PIR sensor detect changes of infrared ray. The sensor can only detect the heat source which is changing or moving from human body. The following items should be noticed. Please confirm the performance and reliability by practical application.

1.1 When detect the heat source besides human body

(1) Pet get into detection area.

(2) In a place exposed directly to sunlight or headlight of automobile.

(3) In a place exposed directly to blow from air-conditioner or heater which make drastic change of temperature in detection area.

1.2 The heat source is hardly detected

(1) In such a place where infrared ray is shaded by glass, propenyl, etc.

(2) The heat source does not move or high-speed move in the detection area.

2. The detection area extended

Even outside the designated detection area, there also exists broad detection area when there is a large temperature difference (above 20°C) between environment and human body.

3. Other usage

3.1 Optical filter of sensor should not be soiled because it may cause failure or malfunction.

3.2 The lens is made by polyethylene. Please avoid stress or impact on the lens, or it will cause performance reduction and work unusually.

3.3 Electronics (above ± 200) should be avoided. Please do not touch terminal by hand.

3.4 Please avoid cleaning the sensor. The cleaning fluid may cause malfunction.

3.5 In order to avoid the interference effect of wires, the shielded wire is recommended and tries to make it short.

3.6 Each Fresnel lens has a fixed focal length. Be careful to adjust its focal length when installing, or the sensitivity will be reduced.

3.7 Be sure the Fresnel lens and the finished casing installed then to test. The sensor's iron shell and pins cannot be exposed.

Note: Senba Sensing reserves the right to make changes, corrections, enhancements, modifications, and improvements to the product specifications at any time without prior notice.